Homework 07: Integer programming models and logic

ISyE 524 — Fall 2026
Due: Monday, November 2, 2026, at 11:59 p.m. (Madison time).

Complete all three problems. Model on/off operating ranges, logical commitments, and minimum set covering. Write your mathematical formulation before implementing it in Julia/JuMP. Use HiGHS and check solver status and solution availability before reading values. Decision-variable definitions are supplied; formulate the objectives and constraints and explain your reasoning.

Submit **one PDF to Gradescope** and mark which pages belong to each problem. **You will not receive full marks unless you mark the pages.**

Getting started and written answers

Follow Assignments and PDF Submission. In VS Code, use **Tasks: Run Task** to run **ISyE 524: Update course repository**, then **ISyE 524: Start an assignment from its template**, entering `hw07`. Open `student-work/hw07/hw07.ipynb`. An existing working copy will not be overwritten.

For **every written part**, type LaTeX mathematics in Markdown or insert a clear photograph/scan of handwritten work. Include the answer and reasoning together and label the subparts; no duplicate transcription is required. Julia/JuMP implementations belong in code cells.

Use `$x_1 \geq 0$` for inline mathematics and `$$x_1+x_2\leq5$$` for display mathematics. Run the Markdown cell to render it.

For handwritten work, edit a Markdown cell in VS Code, drag in a cropped PNG/JPEG, and choose **Insert Image as Attachment**. Alternatively, put the image beside your working notebook under `student-work/hw07/images/` and insert `![My answer](images/problem1.png)`. Keep external images with the notebook. See Images and handwritten work.

Back up `student-work/hw07/` separately; Git does not back up your answers or images. After completing the notebook, export it with the course PDF task, inspect `submissions/hw07.pdf` for readable mathematics, tables, outputs, and images, and submit that PDF to Gradescope.

Computational setup

Use the course Julia environment. All numerical data are supplied below; no separate data files are required. Complete the model and reporting cells. The supplied setup and data cells run without unfinished answers.

AI tools may help debug your own code, explain syntax or error messages, and critique your work. They should not replace constructing your formulation, implementation, or written solution. Complete the required collaboration and LLM-use statement at the end.

```
using JuMP, HiGHS, Printf
import MathOptInterface as MOI
```

1: Steam power with operating minimums

A power plant has three boilers feeding a common steam header and three turbines that draw from it. In one planning period, each boiler and turbine is either off or operates between its stated minimum and maximum. Steam from any boiler may supply any turbine. Steam may be left unused. There are no fixed startup costs.

Boiler i	Minimum steam if on (tons)	Maximum steam (tons)	Steam cost ($/ton)
B1	400	1,000	10
B2	200	900	8
B3	300	800	7

Turbine j	Minimum steam if on (tons)	Maximum steam (tons)	Electricity (kWh/ton)	Processing cost ($/ton)
T1	300	600	4	2
T2	500	800	5	3
T3	600	900	6	4

Use indices $i \in B = \{1, 2, 3\}$ for boilers and $j \in T = \{1, 2, 3\}$ for turbines, with these **supplied decision variables**:

- $x_i \geq 0$: continuous tons of steam produced by boiler i.
- $t_j \geq 0$: continuous tons of steam processed by turbine j.
- $y_i \in \{0, 1\}$: 1 if boiler i operates, 0 otherwise.
- $z_j \in \{0, 1\}$: 1 if turbine j operates, 0 otherwise.

1(a): Formulate and solve

The plant must produce **at least 8,000 kWh**. Formulate a mixed-integer linear program minimizing total boiler and turbine costs. Include constraints that connect each device's operating status to its steam quantity, account for available steam, and meet the electricity requirement. Explain why the operating-range constraints allow a device to be off.

Implement and solve your model. Report the operating devices, steam produced by each boiler, steam processed by each turbine, total electricity, and total cost, with units. Use `x[i]`, `t[j]`, `y[i]`, and `z[j]` for the variables.

Your formulation and reasoning:

[Type your answer here, or insert a clear photograph/scan of handwritten work.]

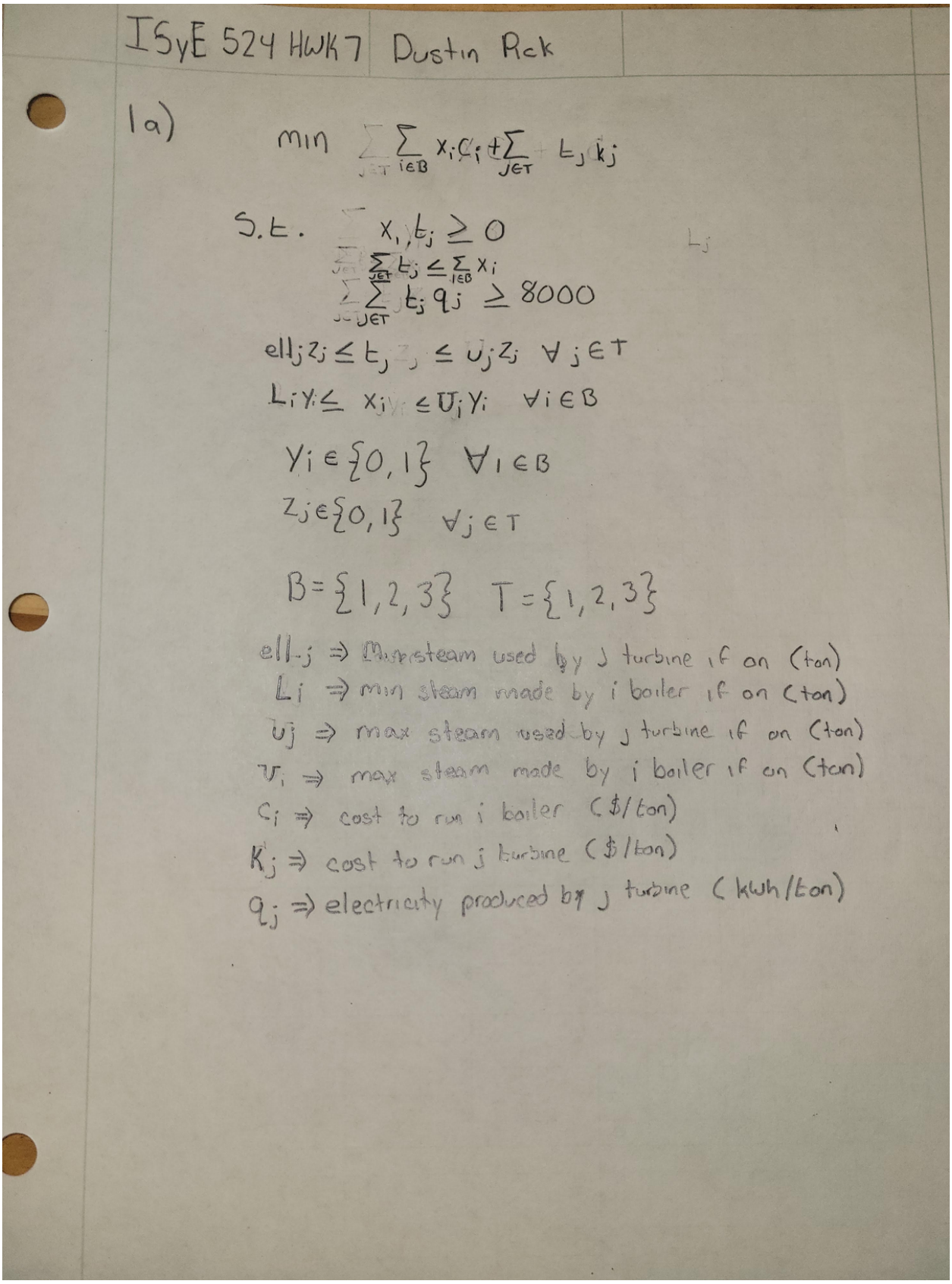


Figure 1: image.png

```
B = 1:3
T = 1:3
L = Dict(zip(B, [400.0, 200.0, 300.0]))
U = Dict(zip(B, [1000.0, 900.0, 800.0]))
c = Dict(zip(B, [10.0, 8.0, 7.0]))
ell = Dict(zip(T, [300.0, 500.0, 600.0]))
u = Dict(zip(T, [600.0, 800.0, 900.0]))
q = Dict(zip(T, [4.0, 5.0, 6.0]))
k = Dict(zip(T, [2.0, 3.0, 4.0]));
```

```
# Build and solve the model for 8,000 kWh.
# You may write a function to reuse the model in 1(b) and 1(c).
# Report device quantities, total electricity, and cost.

m=Model(HiGHS.Optimizer)

@variable(m,z[T], Bin)
@variable(m,y[B], Bin)
@variable(m,x[B]>=0)
@variable(m,t[T]>=0)

@constraint(m, sum(t[j]*q[j] for j in T)>=8000)
@constraint(m,lower_capacity_boiler[i in B], L[i]*y[i]<=x[i])
@constraint(m,upper_capacity_boiler[i in B], x[i]<=U[i]*y[i])
@constraint(m,lower_capacity_turbine[j in T], ell[j]*z[j]<=t[j])
@constraint(m,upper_capacity_turbine[j in T], t[j]<=u[j]*z[j])
@constraint(m, sum(x[i] for i in B)>=sum(t[j] for j in T))

@objective(m, Min, sum(x[i]*c[i] for i in B)+sum(t[j]*k[j] for j in T))

optimize!(m)
if !is_solved_and_feasible(m; allow_local = false)
    error("No optimal primal solution: $(termination_status(m)), $(primal_status
end

println(termination_status(m))

steam_produced=zeros(3)
steam_processed=zeros(3)
electricity_produced=zeros(3)

for i in T
    steam_produced[i]=value(x[i])
    steam_processed[i]=value(t[i])
    electricity_produced[i]=steam_processed[i]*q[i]
end
```

```julia
println("Steam produced by boiler 1: ", steam_produced[1], " tons")
println("Steam produced by boiler 2: ", steam_produced[2], " tons")
println("Steam produced by boiler 3: ", steam_produced[3], " tons")
println("")
println("Steam processed by turbine 1: ", steam_processed[1], " tons")
println("Steam processed by turbine 2: ", steam_processed[2], " tons")
println("Steam processed by turbine 3: ", steam_processed[3], " tons")
println("")
println("Electricity processed by turbine 1: ", electricity_produced[1], " kWh")
println("Electricity processed by turbine 2: ", electricity_produced[2], " kWh")
println("Electricity processed by turbine 3: ", electricity_produced[3], " kWh")
println("")
println("Total electricity produced: ",
 sum(electricity_produced[i] for i in T), " kWh")
println("Minimum cost: ", objective_value(m), " dollars" )
```

```
Running HiGHS 1.15.1 (git hash: 04024d701f): Copyright (c) 2026 under MIT licenc
Includes third-party software components, see THIRD_PARTY_NOTICES.md for full de
Using BLAS: blastrampoline
MIP has 14 rows; 12 cols; 33 nonzeros; 6 integer variables (6 binary)
Coefficient ranges:
  Matrix  [1e+00, 1e+03]
  Cost    [2e+00, 1e+01]
  Bound   [1e+00, 1e+00]
  RHS     [8e+03, 8e+03]
Presolving model
14 rows, 12 cols, 33 nonzeros 0s
10 rows, 10 cols, 25 nonzeros 0s
Presolve reductions: rows 10(-4); columns 10(-2); nonzeros 25(-8)

Solving MIP model with:
   10 rows
   10 cols (4 binary, 0 integer, 0 implied int., 6 continuous, 0 domain fixed)
   25 nonzeros
   Thread count 2 (of 4 threads). Using 1 max workers. Parallel search off

Src: B => Branching; C => Central rounding; F => Feasibility pump; H => Heuristi
     I => Shifting; J => Feasibility jump; L => Sub-MIP; P => Empty MIP; R => Ra
     S => Solve LP; T => Evaluate node; U => Unbounded; X => User solution; Y =>
     Z => ZI Round; l => Trivial lower; p => Trivial point; u => Trivial upper;

        Nodes      |    B&B Tree     |            Objective Bounds
Src  Proc. InQueue |  Leaves   Expl. | BestBound       BestSol              Gap

 J       0       0         0   0.00%   -inf            18800             Large
 R       0       0         0   0.00%   15720           15720             0.00%
         1       0         1 100.00%   15720           15720             0.00%
```

```
Solving report
  Status            Optimal
  Primal bound      15720
  Dual bound        15720
  Gap               0% (tolerance: 0.01%)
  P-D integral      0.000205609646249
  Solution status   feasible
                    15720 (objective)
                    0 (bound viol.)
                    0 (int. viol.)
                    0 (row viol.)
  Timing            0.01
                    0.00 (Presolve)
                    0.01 (Solve)
                    0.00 (Postsolve)
  Max sub-MIP depth 0
  Nodes             1
  Repair LPs        0
  LP iterations     4
                    0 (strong br.)
                    0 (separation)
                    0 (heuristics)
OPTIMAL
Steam produced by boiler 1: 0.0 tons
Steam produced by boiler 2: 620.0 tons
Steam produced by boiler 3: 800.0 tons

Steam processed by turbine 1: 0.0 tons
Steam processed by turbine 2: 520.0 tons
Steam processed by turbine 3: 900.0 tons

Electricity processed by turbine 1: 0.0 kWh
Electricity processed by turbine 2: 2600.0 kWh
Electricity processed by turbine 3: 5400.0 kWh

Total electricity produced: 8000.0 kWh
Minimum cost: 15720.0 dollars
```

Your reported results:

[Type your answer here, or insert a clear photograph/scan of handwritten work.]

The minimum cost for this MIP is 15,720 dollars. To acheive this boiler 1 and turbine 1 are left off and boiler 2 is used to produce 620 tons of steam, boiler 3 is used to produce 800 tons of steam. Turbine 2 processes 520 tons of steam and turbine 3 processes 900 tons of steam to produce 2600 and 5400 kWh of energy respecively. There is a total of 8000kWh of electricity produced.

In this MIP the constraints on upper and lower bounds allow the boilers and turbines to be off becasue when z_j or y_i is zero it forces both the upper and lower bound to be 0, which results in t_j or x_i to be sequeezed to 0 representing its off state.

1(b): Lower electricity demand

Demand falls to **2,000 kWh**. Keep all operating minimums and maximums. Solve the model again. Report only the operating devices and their steam quantities, electricity, and total cost. Explain the change in turbine choice from 1(a).

```
# Solve the same operating-range model for 2,000 kWh and report
# the plan.
mb=Model(HiGHS.Optimizer)

@variable(mb,z[T], Bin)
@variable(mb,y[B], Bin)
@variable(mb,x[B]>=0)
@variable(mb,t[T]>=0)

@constraint(mb, sum(t[j]*q[j] for j in T)>=2000)
@constraint(mb,lower_capacity_boiler[i in B], L[i]*y[i]<=x[i])
@constraint(mb,upper_capacity_boiler[i in B], x[i]<=U[i]*y[i])
@constraint(mb,lower_capacity_turbine[j in T], ell[j]*z[j]<=t[j])
@constraint(mb,upper_capacity_turbine[j in T], t[j]<=u[j]*z[j])
@constraint(mb, sum(x[i] for i in B)>=sum(t[j] for j in T))

@objective(mb, Min, sum(x[i]*c[i] for i in B)+sum(t[j]*k[j] for j in T))

optimize!(mb)
if !is_solved_and_feasible(mb; allow_local = false)
    error("No optimal primal solution: $(termination_status(mb)), $(primal_statu
end

println(termination_status(mb))

steam_produced=zeros(3)
steam_processed=zeros(3)
electricity_produced=zeros(3)

for i in T
    steam_produced[i]=value(x[i])
    steam_processed[i]=value(t[i])
    electricity_produced[i]=steam_processed[i]*q[i]
end

println("Steam produced by boiler 1: ", steam_produced[1], " tons")
println("Steam produced by boiler 2: ", steam_produced[2], " tons")
```

```
println("Steam produced by boiler 3: ", steam_produced[3], " tons")
println("")
println("Steam processed by turbine 1: ", steam_processed[1], " tons")
println("Steam processed by turbine 2: ", steam_processed[2], " tons")
println("Steam processed by turbine 3: ", steam_processed[3], " tons")
println("")
println("Electricity processed by turbine 1: ", electricity_produced[1], " kWh")
println("Electricity processed by turbine 2: ", electricity_produced[2], " kWh")
println("Electricity processed by turbine 3: ", electricity_produced[3], " kWh")
println("")
println("Total electricity produced: ",
 sum(electricity_produced[i] for i in T), " kWh")
println("Minimum cost: ", objective_value(mb), " dollars" )
```

```
Running HiGHS 1.15.1 (git hash: 04024d701f): Copyright (c) 2026 under MIT licenc
Includes third-party software components, see THIRD_PARTY_NOTICES.md for full de
Using BLAS: blastrampoline
MIP has 14 rows; 12 cols; 33 nonzeros; 6 integer variables (6 binary)
Coefficient ranges:
  Matrix  [1e+00, 1e+03]
  Cost    [2e+00, 1e+01]
  Bound   [1e+00, 1e+00]
  RHS     [2e+03, 2e+03]
Presolving model
14 rows, 12 cols, 33 nonzeros 0s
14 rows, 12 cols, 33 nonzeros 0s
Presolve reductions: rows 14(-0); columns 12(-0); nonzeros 33(-0) - Not reduced

Solving MIP model with:
   14 rows
   12 cols (6 binary, 0 integer, 0 implied int., 6 continuous, 0 domain fixed)
   33 nonzeros
   Thread count 2 (of 4 threads). Using 1 max workers. Parallel search off

Src: B => Branching; C => Central rounding; F => Feasibility pump; H => Heuristi
     I => Shifting; J => Feasibility jump; L => Sub-MIP; P => Empty MIP; R => Ra
     S => Solve LP; T => Evaluate node; U => Unbounded; X => User solution; Y =>
     Z => ZI Round; l => Trivial lower; p => Trivial point; u => Trivial upper;

        Nodes      |    B&B Tree     |            Objective Bounds
Src  Proc. InQueue |  Leaves   Expl. | BestBound       BestSol              Gap

 J       0       0         0   0.00%   -inf            6800                Large
         0       0         0   0.00%   3666.666667     6800               46.08%
 C       0       0         0   0.00%   4500            5000               10.00%
 T       0       0         0   0.00%   4500            4500                0.00%
         1       0         1 100.00%   4500            4500                0.00%
```

```
Solving report
  Status            Optimal
  Primal bound      4500
  Dual bound        4500
  Gap               0% (tolerance: 0.01%)
  P-D integral      0.000506254738452
  Solution status   feasible
                    4500 (objective)
                    0 (bound viol.)
                    1.11022302463e-15 (int. viol.)
                    0 (row viol.)
  Timing            0.01
                    0.00 (Presolve)
                    0.01 (Solve)
                    0.00 (Postsolve)
  Max sub-MIP depth 0
  Nodes             1
  Repair LPs        0
  LP iterations     13
                    0 (strong br.)
                    7 (separation)
                    0 (heuristics)
OPTIMAL
Steam produced by boiler 1: 0.0 tons
Steam produced by boiler 2: 0.0 tons
Steam produced by boiler 3: 500.00000000000017 tons

Steam processed by turbine 1: 500.0000000000006 tons
Steam processed by turbine 2: -4.796163466380678e-13 tons
Steam processed by turbine 3: 0.0 tons

Electricity processed by turbine 1: 2000.0000000000025 kWh
Electricity processed by turbine 2: -2.398081733190339e-12 kWh
Electricity processed by turbine 3: 0.0 kWh

Total electricity produced: 2000.0 kWh
Minimum cost: 4500.000000000001 dollars
```

Your answer and reasoning:

[Type your answer here, or insert a clear photograph/scan of handwritten work.]

In this problem it is best to keep boiler 1 and 2 off as well as turbine 2 and 3 off. This will result in producing 500 tons of steam with boiler 3 and processing 500 tons of steam with turbine 1, resulting in a total energy production of 2000 kWh and a cost of 4500 dollars. In this senario it is better to use turbine 1 beacuse all of the capacity can be used and it has the lowest opperating cost whereas in the 8000 kWh demand it was better to use the bigger turbines which require more steam to run but are much more efficent in output vs cost.

1(c): What if operating minimums are ignored?

Keep demand at **2,000 kWh**. Suppose devices may now operate at any nonnegative level up to their maximum when enabled. Remove only the operating-minimum constraints, keeping all other constraints and costs. Solve again. Report the positive steam quantities, electricity, and cost. Compare the cost with 1(b), and identify the operating requirement violated by this plan under the original rules.

```
# Solve the 2,000-kWh case with only the operating minimums
# removed.

# Solve the same operating-range model for 2,000 kWh and report
# the plan.
community=Model(HiGHS.Optimizer)

@variable(community,z[T], Bin)
@variable(community,y[B], Bin)
@variable(community,x[B]>=0)
@variable(community,t[T]>=0)

@constraint(community, sum(t[j]*q[j] for j in T)>=2000)
@constraint(community,lower_capacity_boiler[i in B], 0<=x[i])
@constraint(community,upper_capacity_boiler[i in B], x[i]<=U[i]*y[i])
@constraint(community,lower_capacity_turbine[j in T], 0<=t[j])
@constraint(community,upper_capacity_turbine[j in T], t[j]<=u[j]*z[j])
@constraint(community, sum(x[i] for i in B)>=sum(t[j] for j in T))

@objective(community, Min, sum(x[i]*c[i] for i in B)+sum(t[j]*k[j] for j in T))

optimize!(community)
if !is_solved_and_feasible(community; allow_local = false)
    error("No optimal primal solution: $(termination_status(community)), $(prima
end

println(termination_status(community))

steam_produced=zeros(3)
steam_processed=zeros(3)
electricity_produced=zeros(3)

for i in T
    steam_produced[i]=value(x[i])
    steam_processed[i]=value(t[i])
    electricity_produced[i]=steam_processed[i]*q[i]
end

println("Steam produced by boiler 1: ", steam_produced[1], " tons")
println("Steam produced by boiler 2: ", steam_produced[2], " tons")
```

```
println("Steam produced by boiler 3: ", steam_produced[3], " tons")
println("")
println("Steam processed by turbine 1: ", steam_processed[1], " tons")
println("Steam processed by turbine 2: ", steam_processed[2], " tons")
println("Steam processed by turbine 3: ", steam_processed[3], " tons")
println("")
println("Electricity processed by turbine 1: ", electricity_produced[1], " kWh")
println("Electricity processed by turbine 2: ", electricity_produced[2], " kWh")
println("Electricity processed by turbine 3: ", electricity_produced[3], " kWh")
println("")
println("Total electricity produced: ",
 sum(electricity_produced[i] for i in T), " kWh")
println("Minimum cost: ", objective_value(community), " dollars" )
```

```
Running HiGHS 1.15.1 (git hash: 04024d701f): Copyright (c) 2026 under MIT licenc
Includes third-party software components, see THIRD_PARTY_NOTICES.md for full de
Using BLAS: blastrampoline
MIP has 14 rows; 12 cols; 27 nonzeros; 6 integer variables (6 binary)
Coefficient ranges:
  Matrix  [1e+00, 1e+03]
  Cost    [2e+00, 1e+01]
  Bound   [1e+00, 1e+00]
  RHS     [2e+03, 2e+03]
Presolving model
8 rows, 6 cols, 15 nonzeros 0s
2 rows, 6 cols, 9 nonzeros 0s
Presolve reductions: rows 2(-12); columns 6(-6); nonzeros 9(-18)

Solving MIP model with:
   2 rows
   6 cols (0 binary, 0 integer, 0 implied int., 6 continuous, 0 domain fixed)
   9 nonzeros
   Thread count 2 (of 4 threads). Using 1 max workers. Parallel search off

Src: B => Branching; C => Central rounding; F => Feasibility pump; H => Heuristi
     I => Shifting; J => Feasibility jump; L => Sub-MIP; P => Empty MIP; R => Ra
     S => Solve LP; T => Evaluate node; U => Unbounded; X => User solution; Y =>
     Z => ZI Round; l => Trivial lower; p => Trivial point; u => Trivial upper;

        Nodes      |    B&B Tree     |            Objective Bounds
Src  Proc. InQueue |  Leaves   Expl. | BestBound       BestSol              Gap

 J       0       0         0   0.00%   -inf            8400                Large
 T       0       0         0   0.00%   0               3666.666667       100.00%
         1       0         1 100.00%   3666.666667     3666.666667         0.00%

Solving report
```

```
  Status            Optimal
  Primal bound      3666.66666667
  Dual bound        3666.66666667
  Gap               0% (tolerance: 0.01%)
  P-D integral      0.000185012817383
  Solution status   feasible
                    3666.66666667 (objective)
                    0 (bound viol.)
                    0 (int. viol.)
                    0 (row viol.)
  Timing            0.01
                    0.00 (Presolve)
                    0.01 (Solve)
                    0.00 (Postsolve)
  Max sub-MIP depth 0
  Nodes             1
  Repair LPs        0
  LP iterations     4
                    0 (strong br.)
                    0 (separation)
                    0 (heuristics)
OPTIMAL
Steam produced by boiler 1: 0.0 tons
Steam produced by boiler 2: 0.0 tons
Steam produced by boiler 3: 333.3333333333333 tons

Steam processed by turbine 1: 0.0 tons
Steam processed by turbine 2: 0.0 tons
Steam processed by turbine 3: 333.33333333333337 tons

Electricity processed by turbine 1: 0.0 kWh
Electricity processed by turbine 2: 0.0 kWh
Electricity processed by turbine 3: 2000.0000000000002 kWh

Total electricity produced: 2000.0000000000002 kWh
Minimum cost: 3666.6666666666665 dollars
```

Your answer and reasoning:

[Type your answer here, or insert a clear photograph/scan of handwritten work.]

If minimums opperating constrints are removed then it is best to leave boiler 1 and 2 off as well as turbine 1 and 2. Then 334 tons of steam will be produced by boiler 3 and processed by turbine 3 to resut in 2000 kWh for a cost of 3667 dollars. This produces a lower cost than having minimum opperating requrements becasue it can use the most efficent turbine in terms of ouput per cost but it violates the minimum posed in the original rules becasue it only used 333 tons of capacity of turbine 3 when its minimum is 600 tons.

2: Community projects with logical commitments

A community organization has **$24,000** for projects. Each project must be undertaken completely or omitted. Costs below are in **thousands of dollars**. Impact points are a planning score that adds across projects. Field-team hours are the total hours required by each selected project.

Project p	Description	Cost c_p	Impact b_p	Hours h_p
A	Community garden	7	13	6
B	Food-truck pop-up	6	12	5
C	Volunteer training	5	9	4
D	Tool library	4	8	4
E	Storage upgrade	3	5	2
F	Compost hub	4	7	3
G	Soil workshop	2	4	1
H	Nutrition class	5	10	5

The organization has five operating commitments:

1. The garden requires volunteer training or the tool library, or both.
2. The food-truck pop-up and tool library cannot both be undertaken because they need the same trailer.
3. If both volunteer training and the tool library are undertaken, the storage upgrade must also be undertaken. The upgrade may be selected independently.
4. The compost hub and soil workshop must be undertaken together or both omitted.
5. Exactly one of the food-truck pop-up and nutrition class must be selected.

Selecting both the garden and nutrition class earns **four additional impact points** for their combined program. This bonus is awarded once, and only when both are selected. There is no other limit on total hours in parts (a)–(b).

Use these **supplied decision variables**:

- $x_p \in \{0,1\}$ for $p \in P = \{A, B, C, D, E, F, G, H\}$: 1 if project p is selected, 0 otherwise.
- $w \in \{0,1\}$: 1 if and only if both A and H are selected.

2(a): Formulate the logical selection model

Formulate a binary linear program maximizing total impact, including the joint-program bonus, subject to the budget and all five commitments. Ensure the constraints give w its stated meaning in every feasible solution. Label the constraints corresponding to commitments 1–5.

Explain why commitment 1 permits both supporting projects and why commitment 3 permits the storage upgrade even when the training and tool library are not both selected. Do not use products of decision variables.

Your formulation and reasoning:

[Type your answer here, or insert a clear photograph/scan of handwritten work.]

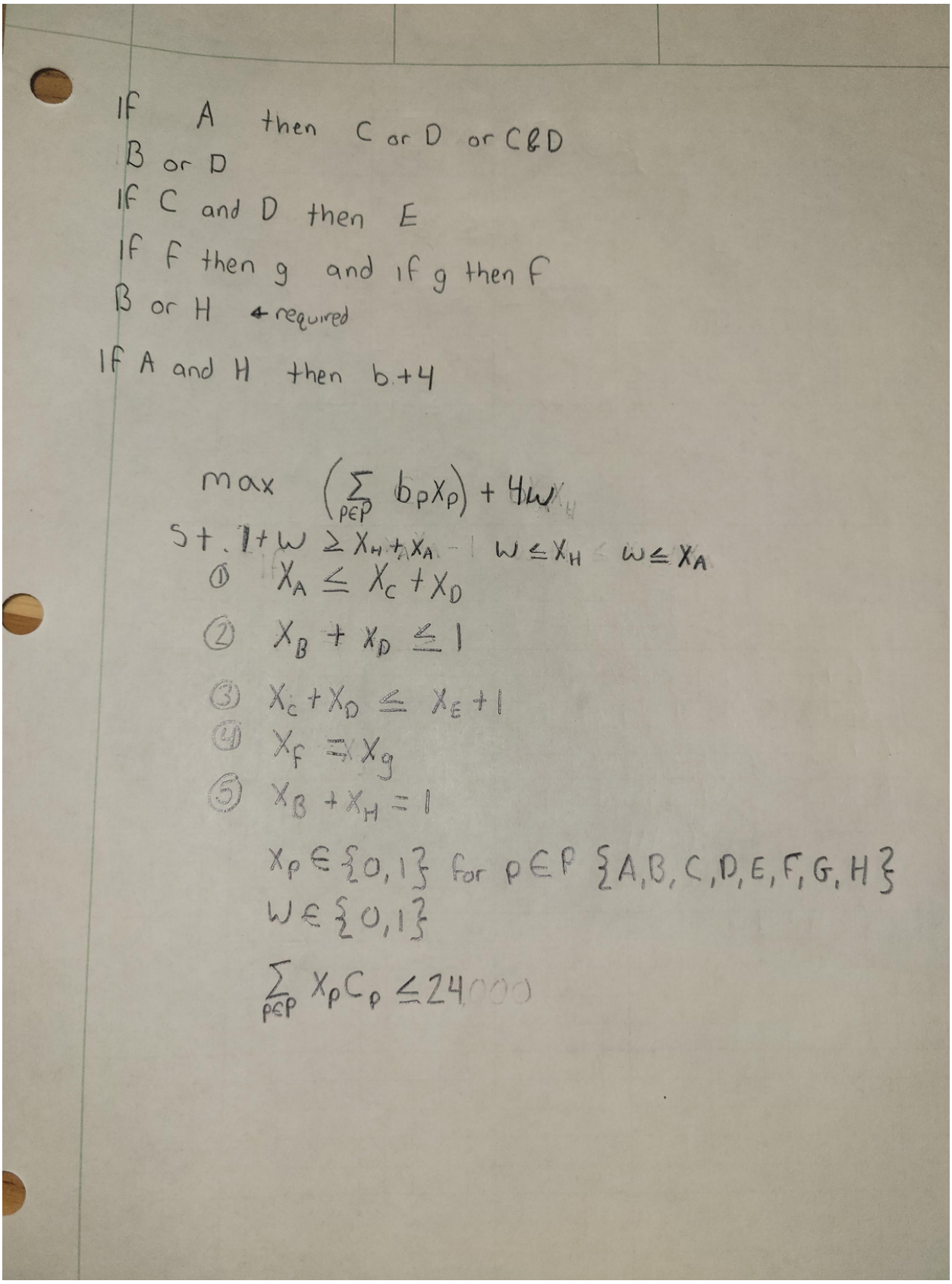


Figure 2: image.png

The constraint $ X_A X_C + X_D $ ensures if a is chosen then C, D, or C and D must be chosen becasue if $X_A = 1$ then $X_C = 1$ and $X_D = 1$ results in $1 \leq 2$ which makes the stament true. $X_C = 0$ and $X_D = 1$ results in $1 \leq 1$ which makes the stament true. $X_C = 1$ and $X_D = 0$ results in $1 \leq 1$ which makes the stament true. But $X_C = 0$ and $X_D = 0$ results in $1 \leq 0$ which makes the stament false.

The constraint $X_C + X_D \leq X_E + 1$ allows E to be selected when C and D are not both selected becasue it results in $X_C = 0$ and $X_D = 0$ $X_E = 1$ results in the constraint holding becasue $0 \leq 2$. It also allows one of D or C to be selected as well as E becasue that will result in $1 \leq 2$ the only thing it doesnt allow is E to be 0 and D and C are 1 which would result in $2 \leq 1$ which would make the constraint false.

```
P = [:A, :B, :C, :D, :E, :F, :G, :H]
c_project = Dict(zip(P, [7, 6, 5, 4, 3, 4, 2, 5]))
b_project = Dict(zip(P, [13, 12, 9, 8, 5, 7, 4, 10]))
h_project = Dict(zip(P, [6, 5, 4, 4, 2, 3, 1, 5]))
budget = 24;  # thousands of dollars
```

2(b): Solve and interpret

Implement and solve the model from 2(a). Use `x[p]` and `w` for the variables. Report selected projects, project spending in dollars, total field-team hours, and total impact including any bonus. State which projects earn the bonus, if it is awarded.

```
# Build and solve the project-selection model.
# Report selected projects, spending, hours, and impact.

community=Model(HiGHS.Optimizer)

@variable(community,x[P], Bin)
@variable(community,w, Bin)


@constraint(community, sum(x[p]*c_project[p] for p in P)<= budget)
@constraint(community,1+w>=x[:H]+x[:A])
@constraint(community, w <= x[:H])
@constraint(community, w <= x[:A])
@constraint(community,x[:A] <= x[:C]+x[:D])
@constraint(community, x[:D]+x[:B]<=1)
@constraint(community, x[:C]+x[:D]<= x[:E]+1)
@constraint(community, x[:F]==x[:G])
@constraint(community, x[:B]+x[:H]==1)

@objective(community, Max, sum(b_project[p]*x[p] for p in P)+4*w)

optimize!(community)
```

```
if !is_solved_and_feasible(community; allow_local = false)
    error("No optimal primal solution: $(termination_status(community)), $(prima
end

println(termination_status(community))

println("Bonus value Status: ", value.(w))
println("Project A status : ", value.(x[:A]))
println("Project B status : ", value.(x[:B]))
println("Project C status : ", value.(x[:C]))
println("Project D status : ", value.(x[:D]))
println("Project E status : ", value.(x[:E]))
println("Project F status : ", value.(x[:F]))
println("Project G status : ", value.(x[:G]))
println("Project H status : ", value.(x[:H]))
println("")
println("Total Cost: ",sum(value.(x[p]*c_project[p] for p in P)), " dollars")
println("")
println("Feild Team hours: ",sum(value.(x[p]*h_project[p] for p in P)), " hours"
println("")
println("Maximum  impact: ", objective_value(community) )
```

```
Running HiGHS 1.15.1 (git hash: 04024d701f): Copyright (c) 2026 under MIT licenc
Includes third-party software components, see THIRD_PARTY_NOTICES.md for full de
Using BLAS: blastrampoline
MIP has 9 rows; 9 cols; 27 nonzeros; 9 integer variables (9 binary)
Coefficient ranges:
  Matrix  [1e+00, 7e+00]
  Cost    [4e+00, 1e+01]
  Bound   [1e+00, 1e+00]
  RHS     [1e+00, 2e+01]
Presolving model
7 rows, 7 cols, 21 nonzeros 0s
7 rows, 7 cols, 21 nonzeros 0s
Presolve reductions: rows 7(-2); columns 7(-2); nonzeros 21(-6)
Objective function is integral with scale 1

Solving MIP model with:
   7 rows
   7 cols (7 binary, 0 integer, 0 implied int., 0 continuous, 0 domain fixed)
   21 nonzeros
   Thread count 2 (of 4 threads). Using 1 max workers. Parallel search off

Src: B => Branching; C => Central rounding; F => Feasibility pump; H => Heuristi
     I => Shifting; J => Feasibility jump; L => Sub-MIP; P => Empty MIP; R => Ra
     S => Solve LP; T => Evaluate node; U => Unbounded; X => User solution; Y =>
     Z => ZI Round; l => Trivial lower; p => Trivial point; u => Trivial upper;
```

```
        Nodes      |    B&B Tree     |            Objective Bounds
Src  Proc. InQueue |  Leaves   Expl. | BestBound       BestSol              Gap

 z       0       0         0   0.00%   inf             12                 Large
 J       0       0         0   0.00%   inf             23                 Large
 R       0       0         0 100.00%   49.5            49                 1.02%
         1       0         1 100.00%   49              49                 0.00%

Solving report
  Status            Optimal
  Primal bound      49
  Dual bound        49
  Gap               0% (tolerance: 0.01%)
  P-D integral      0.000266794621045
  Solution status   feasible
                    49 (objective)
                    0 (bound viol.)
                    0 (int. viol.)
                    0 (row viol.)
  Timing            0.01
                    0.00 (Presolve)
                    0.01 (Solve)
                    0.00 (Postsolve)
  Max sub-MIP depth 0
  Nodes             1
  Repair LPs        0
  LP iterations     3
                    0 (strong br.)
                    0 (separation)
                    0 (heuristics)
OPTIMAL
Bonus value Status: 1.0
Project A status : 1.0
Project B status : 0.0
Project C status : 1.0
Project D status : 1.0
Project E status : 1.0
Project F status : 0.0
Project G status : 0.0
Project H status : 1.0

Total Cost: 24.0 dollars

Feild Team hours: 21.0 hours

Maximum  impact: 49.0
```

Your results and interpretation:

[Type your answer here, or insert a clear photograph/scan of handwritten work.]

The maximum impact possible from these programs is 49. This is done by selecting project A,C,D,E,and H. Becasue A and H are selected together the bonus impact is earned. The projects cost 24000 dollars in total and ther is 21 feild team hours used.

2(c): A workload-triggered supervisor fee

Now a supervisor must be hired if total field-team hours are **17 or more**. Below 17 hours, no supervisor is hired. The supervisor costs **$3,000**, which counts against the **same $24,000 budget**; it adds no impact points. Keep all five operating commitments and the joint-program bonus.

Use the additional **supplied binary variable** v, equal to 1 if and only if total field-team hours are at least 17. Write the added or changed linear constraints and justify finite constants using the data. Explain why the integer hour total permits an exact distinction between fewer than 17 hours and at least 17 hours.

Implement and solve the revised model, using v for the new variable. Report selected projects, project spending, supervisor cost, combined spending, hours, and impact. Explain the change from 2(b).

Your formulation and reasoning:

[Type your answer here, or insert a clear photograph/scan of handwritten work.]

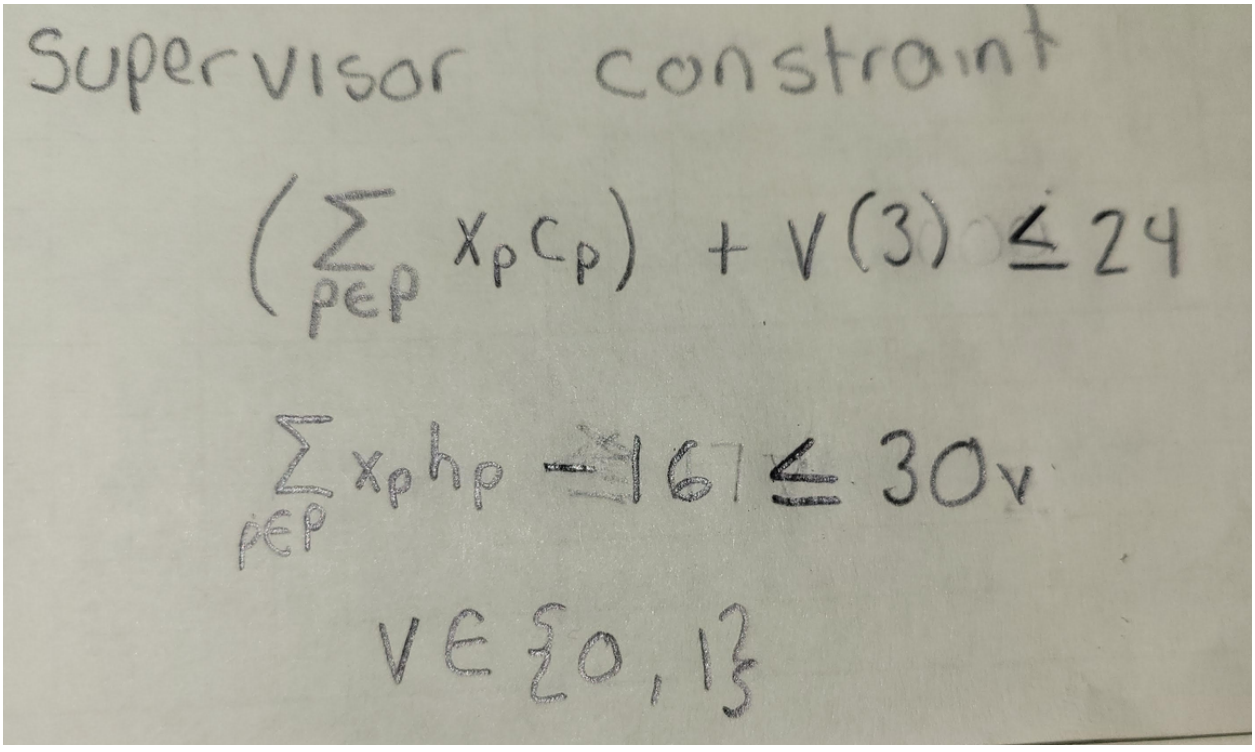


Figure 3: image-2.png

A new term is added to the cost constraint which charges the budged 3000 dollars if the supervisor is needed because when the supervisor is needed v=1 and the cost is active but when the supervisor isnt needed the v=0 and the cost is not added.

The total hours -16 $ 30*z$ constraint makes sure that the v is 1 if hours is 17 or greater. This is becasue any value 16 or lower will result in a value less than 0 so v can remain 0. Any value 17 or greater will be positive and require v to be 1. 30 was chosen as the big M because it is the total possible time if all projects were to be selected, thus it is a safe upper limit. Along with this the fact that all Xp values and hp values are integers means that fewer than 17 results in less than or equal to 16 and at least 17 is the same as greater than or equal to 17.

```
# Add the exact workload indicator and include the supervisor in
# the budget.
# Solve and report the revised selection, costs, hours, and
# impact.

communityc=Model(HiGHS.Optimizer)

@variable(communityc,x[P], Bin)
@variable(communityc,w, Bin)
@variable(communityc,v, Bin)

@constraint(communityc, sum(x[p]*c_project[p] for p in P)+v*3<= budget)
@constraint(communityc,1+w>=x[:H]+x[:A])
@constraint(communityc, w <= x[:H])
@constraint(communityc, w <= x[:A])
@constraint(communityc,x[:A] <= x[:C]+x[:D])
@constraint(communityc, x[:D]+x[:B]<=1)
@constraint(communityc, x[:C]+x[:D]<= x[:E]+1)
@constraint(communityc, x[:F]==x[:G])
@constraint(communityc, x[:B]+x[:H]==1)
@constraint(communityc, sum(h_project[p]*x[p] for p in P)-16 <= 30*v)

@objective(communityc, Max, sum(b_project[p]*x[p] for p in P)+4*w)

optimize!(communityc)
if !is_solved_and_feasible(communityc; allow_local = false)
    error("No optimal primal solution: $(termination_status(communityc)), $(prim
end

println(termination_status(communityc))

println("Bonus value Status: ", value.(w))
println("Supervisor Status: ", value.(v))
println("Supervisor Cost :", value.(v)*3000, " dollars")
println("")
println("Project A status : ", value.(x[:A]))
println("Project B status : ", value.(x[:B]))
```

```
println("Project C status : ", value.(x[:C]))
println("Project D status : ", value.(x[:D]))
println("Project E status : ", value.(x[:E]))
println("Project F status : ", value.(x[:F]))
println("Project G status : ", value.(x[:G]))
println("Project H status : ", value.(x[:H]))
println("")
println("Total Cost: ",sum(value(x[p])*c_project[p] for p in P)+value(v)*3,
 " dollars")
println("")
println("Feild Team hours: ",sum(value(x[p])*h_project[p] for p in P), " hours")
println("")
println("Maximum  impact: ", objective_value(communityc) )
```

```
Running HiGHS 1.15.1 (git hash: 04024d701f): Copyright (c) 2026 under MIT licenc
Includes third-party software components, see THIRD_PARTY_NOTICES.md for full de
Using BLAS: blastrampoline
MIP has 10 rows; 10 cols; 37 nonzeros; 10 integer variables (10 binary)
Coefficient ranges:
  Matrix  [1e+00, 3e+01]
  Cost    [4e+00, 1e+01]
  Bound   [1e+00, 1e+00]
  RHS     [1e+00, 2e+01]
Presolving model
8 rows, 8 cols, 28 nonzeros 0s
8 rows, 8 cols, 28 nonzeros 0s
Presolve reductions: rows 8(-2); columns 8(-2); nonzeros 28(-9)
Objective function is integral with scale 1

Solving MIP model with:
   8 rows
   8 cols (8 binary, 0 integer, 0 implied int., 0 continuous, 0 domain fixed)
   28 nonzeros
   Thread count 2 (of 4 threads). Using 1 max workers. Parallel search off

Src: B => Branching; C => Central rounding; F => Feasibility pump; H => Heuristi
     I => Shifting; J => Feasibility jump; L => Sub-MIP; P => Empty MIP; R => Ra
     S => Solve LP; T => Evaluate node; U => Unbounded; X => User solution; Y =>
     Z => ZI Round; l => Trivial lower; p => Trivial point; u => Trivial upper;

        Nodes      |    B&B Tree     |            Objective Bounds
Src  Proc. InQueue |  Leaves   Expl. | BestBound       BestSol              Gap

 z       0       0         0   0.00%   inf             12                 Large
 J       0       0         0   0.00%   inf             23                 Large
 R       0       0         0   0.00%   47.14285714     35                34.69%
 C       0       0         0   0.00%   44.17948718     40                10.45%
```

```
 L       0       0         0   0.00%   44.17948718     41                7.75%
         1       0         1 100.00%   41              41                0.00%

Solving report
  Status            Optimal
  Primal bound      41
  Dual bound        41
  Gap               0% (tolerance: 0.01%)
  P-D integral      0.00076587007057
  Solution status   feasible
                    41 (objective)
                    0 (bound viol.)
                    0 (int. viol.)
                    0 (row viol.)
  Timing            0.01
                    0.00 (Presolve)
                        MIP    time [calls] = 0.00 [1]
                        subMIP time [calls] = 0.00 [2]
                    0.01 (Solve)
                    0.00 (Postsolve)
                        MIP    time [calls] = 0.00 [1]
                        subMIP time [calls] = 0.00 [2]
  Max sub-MIP depth 2
  Nodes             1
  Repair LPs        0
  LP iterations     26
                    4 (strong br.)
                    11 (separation)
                    0 (heuristics)
OPTIMAL
Bonus value Status: 1.0
Supervisor Status: 1.0
Supervisor Cost :3000.0 dollars

Project A status : 1.0
Project B status : 0.0
Project C status : 1.0
Project D status : 0.0
Project E status : 1.0
Project F status : 0.0
Project G status : 0.0
Project H status : 1.0

Total Cost: 23.0 dollars

Feild Team hours: 17.0 hours

Maximum  impact: 41.0
```

Your results and explanation:

[Type your answer here, or insert a clear photograph/scan of handwritten work.]

With the added constraints from the supervisor the projects selected are A,C,E,and H. Becasue A and H are selected the bonus impact is earned. The maximum impact possible from this new set of constraints is 41, with a total cost of 23000 dollars. The supervisor is needed and costs 3000 dollars becasue the feild team uses 17 total hours. This is different from the soloution in b because once the feild hours passes 17 hours the 3000 dollar supervior charge is induced and there isn't enough money to do another project. Thus one less project is done in this plan then in the plan in part b.

3: Auror posts and minimum set covering

The wizarding world has eight districts. The Ministry of Magic wants to station auror posts so every district can be reached from at least one post within **two seconds**. A post can protect its own district. There is at most one post per district, and all posts have the same cost. There is no capacity limit: one post can protect several districts.

For each district i, the table gives N_i, the candidate post locations that can protect it. These coverage sets are supplied; no travel-time calculation is required.

District i	Eligible post locations N_i
1	1, 5
2	2, 7
3	3, 4, 5
4	3, 4, 6
5	1, 3, 5, 6, 7
6	4, 5, 6, 8
7	2, 5, 7, 8
8	6, 7, 8

Use the **supplied decision variable** $x_j \in \{0, 1\}$ for $j \in J = \{1, \dots, 8\}$, equal to 1 if a post is opened at district j, and 0 otherwise.

3(a): Formulate the covering model

Formulate an integer linear program minimizing the number of posts while protecting all eight districts. Explain why the coverage requirement for a district is an inequality rather than an equality.

Your formulation and reasoning:

[Type your answer here, or insert a clear photograph/scan of handwritten work.]

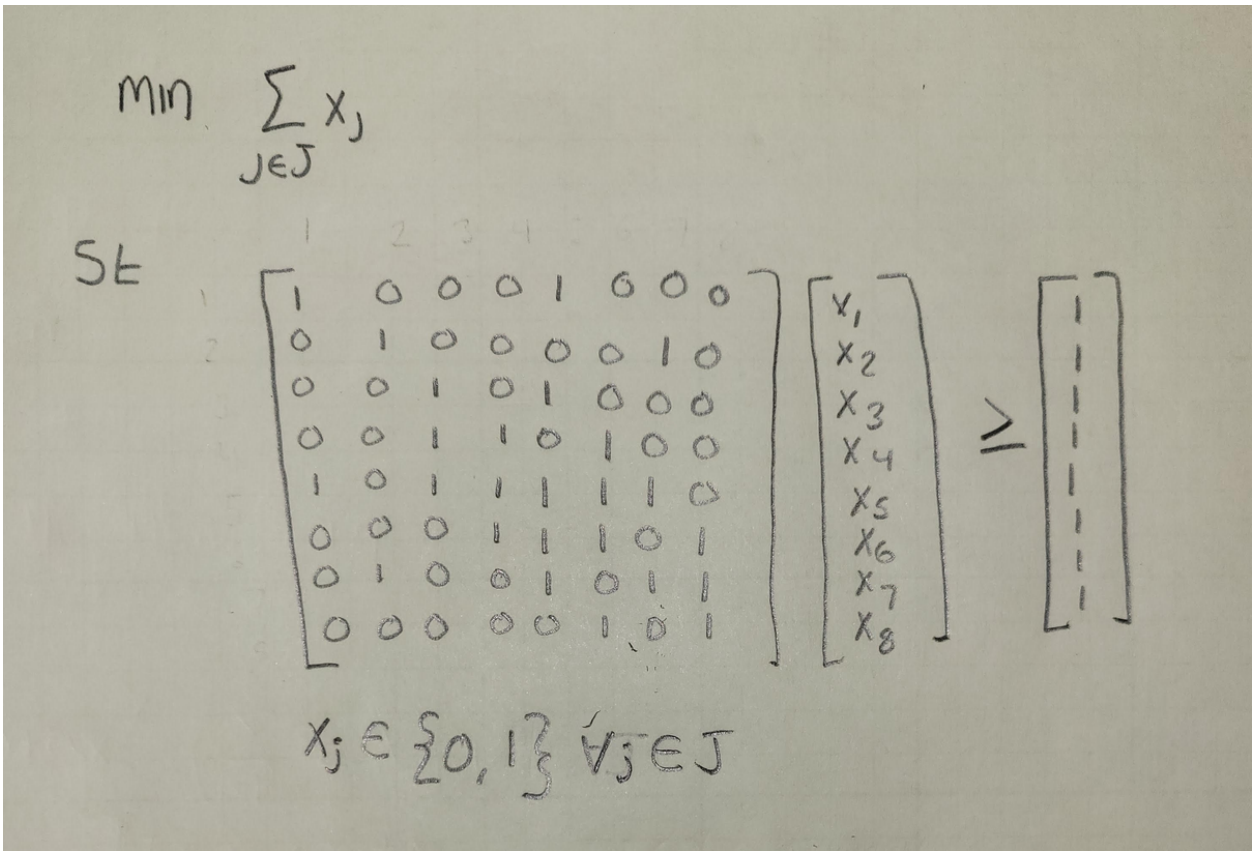


Figure 4: image.png

The constraint must be an inequality in this problem because we need to make sure we cover all locations and it is acceptable if one is covered more than once. If we set it to an equality we would be forcing each post to be covered exactly once which may not be possible and may make the problem infeasable.

```
I = 1:8
J = 1:8
N = Dict(1 => [1, 5],
         2 => [2, 7],
         3 => [3, 4, 5],
         4 => [3, 4, 6],
         5 => [1, 3, 5, 6, 7],
         6 => [4, 5, 6, 8],
         7 => [2, 5, 7, 8],
         8 => [6, 7, 8]);
```

3(b): Locate posts and verify coverage

Implement and solve your model, using `x[j]` for the binary decisions. Report the minimum number of posts and their locations. For every district, list the selected posts that protect it, verifying that

none is unprotected. Any optimal set of locations is acceptable. No column generation or routing algorithm is required.

```
# Build and solve the minimum set-covering model.
# Report open locations and, for each district, the open posts
# protecting it.

a=Model(HiGHS.Optimizer)

@variable(a,x[J], Bin)


n = length(N)
N2 = zeros(Int, n, n)

for (row, cols) in N
    N2[row, cols] .= 1
end

b= [1
    1
    1
    1
    1
    1
    1
    1]

@constraint(a, N2*x .>=b)

@objective(a, Min, sum(x[j] for j in J))

optimize!(a)
if !is_solved_and_feasible(a; allow_local = false)
    error("No optimal primal solution: $(termination_status(a)), $(primal_status
end

println(termination_status(a))

println("number of posts needed: ",objective_value(a))
println("number of each post:")
for i in 1:length(x)
    println("post ", i, " status ",value.(x[i]))
end
```

```
Running HiGHS 1.15.1 (git hash: 04024d701f): Copyright (c) 2026 under MIT licenc
Includes third-party software components, see THIRD_PARTY_NOTICES.md for full de
```

```
Using BLAS: blastrampoline
MIP has 8 rows; 8 cols; 26 nonzeros; 8 integer variables (8 binary)
Coefficient ranges:
  Matrix  [1e+00, 1e+00]
  Cost    [1e+00, 1e+00]
  Bound   [1e+00, 1e+00]
  RHS     [1e+00, 1e+00]
Presolving model
8 rows, 8 cols, 26 nonzeros 0s
0 rows, 0 cols, 0 nonzeros 0s
Presolve reductions: rows 0(-8); columns 0(-8); nonzeros 0(-26) - Reduced to emp
Presolve: Optimal

Src: B => Branching; C => Central rounding; F => Feasibility pump; H => Heuristi
     I => Shifting; J => Feasibility jump; L => Sub-MIP; P => Empty MIP; R => Ra
     S => Solve LP; T => Evaluate node; U => Unbounded; X => User solution; Y =>
     Z => ZI Round; l => Trivial lower; p => Trivial point; u => Trivial upper;

        Nodes      |    B&B Tree     |            Objective Bounds
Src  Proc. InQueue |  Leaves   Expl. | BestBound       BestSol              Gap

         0       0         0   0.00%   3               3                 0.00%

Solving report
  Status            Optimal
  Primal bound      3
  Dual bound        3
  Gap               0% (tolerance: 0.01%)
  P-D integral      0
  Solution status   feasible
                    3 (objective)
                    0 (bound viol.)
                    0 (int. viol.)
                    0 (row viol.)
  Timing            0.00
                    0.00 (Presolve)
                    0.00 (Solve)
                    0.00 (Postsolve)
  Max sub-MIP depth 0
  Nodes             0
  Repair LPs        0
  LP iterations     0
OPTIMAL
number of posts needed: 3.0
number of each post:
post 1 status 0.0
post 2 status 0.0
post 3 status 1.0
```

```
post 4 status 0.0
post 5 status 1.0
post 6 status 0.0
post 7 status 1.0
post 8 status 0.0
```

Your results and interpretation:

[Type your answer here, or insert a clear photograph/scan of handwritten work.]

The minimum number of posts is 3, being post 3 which protects posts 3,4,and 5 post 5 which protects posts 1, 3, 5, 6, and 7 and post 7 which protects 2, 5, 7 and 8.

Post	Protector
1	5
2	7
3	3,5
4	3
5	3,5,7
6	5
7	5,7
8	7

Collaboration and LLM-use statement — required

Complete this statement as the **very last part of your homework**. Be specific enough that a reader can understand what work you did yourself and how others or LLMs helped you learn.

- **People:** Give the name of every person you collaborated with, identify the problem/subpart you discussed, and explain the nature of the collaboration. If you worked alone, explicitly state that you did not collaborate with anyone.
- **LLMs/tools:** Name every LLM or AI assistant you used and identify the model/version as precisely as available. If the interface did not show a model/version, say so. If you used none, explicitly state that you did not use an LLM or AI assistant.
- **Specific assistance and learning:** For each tool, identify the problem/subpart, describe what you asked it to help with, what suggestions you used or rejected, and how the interaction helped you understand the mathematics or implementation. Explain how you checked the suggestions. "I used ChatGPT for help" is not sufficiently specific.

Disclosure does not replace the requirement that your submitted formulation, implementation, and explanation be your own work and that you understand them.

People and collaboration:

[Names, problem/subparts, and what you discussed; or explicitly state that you worked alone.]

I worked alone

LLMs/tools and models:

[Tool names and models/versions, or explicitly state that you used none.]

I utilized Gemini 3.6.

How I used the assistance, what I learned, and how I checked it:

[Specific details for each tool and relevant problem/subpart; state not applicable if you used none.]

I used it to help debug my code in problem 1a to help debug and explain the errors I was having that were ultimately caused by accidently having a extra bracket in my [j in T] (it was [j]in T]).

I used in in problem 2b as well to help explain the difference between value() and value.(). I learned that value.() is needed for any values that are arrays and value. works on scalars